Thanh Le-Cong

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thanhlecongg homepage

Education

Aug 2016 – July 2021

Hanoi University of Science and Technology (HUST)
BE (Honours) in Information Technology

- Thesis topic: Development of Automated Patch Validation for Program Repair
- Selected to Talented Engineering Program (30 out of 1000 students in our cohorts)
- GPA: 3.2/4.0
- Thesis: 9.7/10.0

Research Interests

My research interests lie at the intersection of *software engineering*, *artificial intelligence*. I am currently focusing on **automated debugging for software and artificial intelligent system** with a focus on specific topics such as automated bug identification, localization and repair (C2, C4, C6, U1, U2)*, reducing false postives in static analysis (C5)* and analyzing deep neural network (C3)*.

**C2, C3, C4, C5, C6, U1, U2 denotes my works which is listed on publication section*

Publications

C: Conference, J: Journal, U: Under Submission
Note * denote denotes that I am the co-first author.

1. **[C6] VulCurator: A Vulnerability-Fixing Commit Detector**, by Truong-Giang Nguyen, Thanh Le Cong, Kang Hong Jin, Xuan-Bach D. Le, David Lo; *accepted at the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Demonstration Track, ACM, 2022 (5 pages, Acceptance Rate = 56.5%)*
2. **[C5] AutoPruner: Transformer-based Call Graph Pruning** by Thanh Le Cong, Kang Hong Jin, Truong-Giang Nguyen, Stefanus Agus Haryono, David Lo, Xuan-Bach D. Le, Quyet-Thang Huynh; *accepted at the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), Research Track, ACM, 2022 (10 pages, Acceptance Rate = 22%)*
3. **[C4] FFL: Fine grained Fault Localization for Student Programs via Syntactic and Semantic Reasoning**, by Thanh-Dat Nguyen, Thanh Le Cong, Xuan Bach D. Le, Duc Minh Luong, David Lo, Quyet-Thang Huynh *accepted at the 38th IEEE International Conference on Software Maintenance and Evolution (ICSME), Research Track, IEEE, 2022 (10 pages, Acceptance Rate = 23%)*
4. **[C3] Towards the Analysis of Graph Neural Network**, by Thanh-Dat Nguyen, Thanh Le Cong*, ThanhVu H. Nguyen, Xuan Bach D. Le, Huynh Quyet Thang, *in the 2022 IEEE/ACM 44th International Conference on Software Engineering: New Ideas and Emerging Results (ICSE-NIER), IEEE, 2022 (5 pages, Acceptance Rate = 26.8%)*
5. **[C2] Usability and Aesthetic: Better Together for Automated Repair of Web Pages**, by Thanh Le Cong, Xuan Bach D. Le, Quyet Thang Huynh, Phi Le Nguyen, *in Proceeding of the 32nd IEEE International Symposium on Software Reliability Engineering (ISSRE), Research Track. IEEE, 2021 (10 pages, Acceptance Rate = 27.5%)*.
6. **[J1] A Multifactorial Optimization Paradigm for Linkage Tree Genetic Algorithm** by Thi Thanh Binh Huynh, Dinh Thanh Pham, Ba Trung Tran, Cong Thanh Le, Minh Hai Phong Le, Ananthram Swami, Thu Lam Bui, *in Information Sciences 540 (2020): 325-344 (19 pages, Impact Factor = 6.795)*
7. **[C1] SAFL: A Self-Attention Scene Text Recognizer with Focal Loss**, by Bao Hieu Tran, Thanh Le Cong*, Huu Manh Nguyen, Duc Anh Le, Thanh Hung Nguyen, Phi Le Nguyen, *in Proceedings of the 19th IEEE International Conference On Machine Learning And Applications (ICMLA), IEEE, 2020 (6 pages, Acceptance Rate = 25.1%)*

Works Under Submission

1. **[U1] Automated Patch Validation via program invariants**, by Thanh Le Cong, Xuan Bach D. Le, David Lo, Quyet Thang Huynh, Quang Huy Bui, Nhat Hoa Tran *(Major Revision, IEEE Transaction on Software Engineering (TSE))*
2. **[U2] MiDas: Multi-granularity Model for Vulnerability-fixing Commit Classification**, by Truong-Giang Nguyen, Thanh Le Cong, Kang Hong Jin, Bowen Xu, Jiayuan Zhou, Xin Xia, Ahmed E. Hassan Xuan Bach D. Le, David Lo; *(Revise and Resubmit, IEEE Transaction on Software Engineering (TSE))*

Research Experience

Nov 2021 - present

Software Analytics Research Group, Singapore Management University
Researcher Engineer

- Research on automted debugging for Software and AI systems under the supervision of Prof. David Lo (SMU) and Dr. Le Dinh Xuan Bach (University of Melbourne)
- Current projects: (1) Vulnerability detection for machine learning frameworks; (2) Property Inference for Deep Neural Networks and (3) Bug localization and repair;
- Result: Published 3 paper including 3 A*-ranked conference paper and 1 A-ranked conference paper; and submitted 2 manusripts for TSE journals.

June 2017 - Dec 2021

Hanoi University of Science and Technology
Undergraduate Research Assistant

- Led and collaborated with professors on 6+ research projects on various topics including software engineering, computer vision and evolutionary algorithm.
- Published 4 papers including 1 A-ranked papers (first author), 1 C-ranked paper (first author) and 1 journal paper (co-author)

July 2019

Big Data Visualisation Lab, Griffith University
Undergraduate Research Intern

- Develop a novel node embedding techiniques based on community detection;
- Result: published a preprint, which is available online in Openreview (`https://openreview.net/forum?id=Byl3K2VtwB`).

Skills

Programmmig Languages	Python, Java, Rust, C/C++
Deep Learning Frameworks	Pytorch, Tensorflows
Machine Learning Libraries	Sklearn, Numpy, Pandas, Networkx, DGL, OpenCV
Operating Systems	Linux, Windows
Database Management	MS SQLServer, MySQL
Editor	Latex, MS Office

Language Skills

- Vietamese (mother tongue)
- English (IELTS 6.0)

Awards

** Certifications for the awards is available online [1]

[1] `https://drive.google.com/drive/folders/1weOY7Qp9jcjiRXvNOFwpQkUHhEzPTrUH?usp=sharing`

2021 **[A1] Best presentation award in thesis defense**
Hanoi University of Science and Technology

2021 **[A2] University scholarship for academic excellency (Top 1%)**
Hanoi University of Science and Technology

2020 **[A3] Winner in Contest of "Innovative Information Technology Products" in 2020**
Hanoi University of Science and Technology

2020 **[A4] Runner-up Prize of the Scientific Research Student Award**
Hanoi University of Science and Technology

2018 **[A5] Winner in IEEE WCCI competition on Evolutionary Multi-task optimization**
IEEE

2017 **[A6] First Prize in Vietnam Mathematics Olympiad for University Students**
Vietnam Mathematical Society

2016 **[A7] Silver Meldal in the 38th International Mathematic Tournament of Cities in Hanoi**
Vietnam National University

2016 **[A8] Selected Participant (Top 30 nationwide) in Vietnam Team Selection Test for Internation Mathematical Olympiad**
Vietnam Ministry of Education and Training

2016 **[A9] Second Prize of Mathematics in National Competition for Excellent High School Students**
Vietnam Ministry of Education and Training

2014 **[A10] Scholarship of the National Key Program on Mathematics Development**
Vietnam Ministry of Education and Training

References

Dr. Bach Le
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University of Melbourne
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Singapore Management University
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Dr. Huynh Quyet Thang
Associate Professor, President
Hanoi University of Science and Technology
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